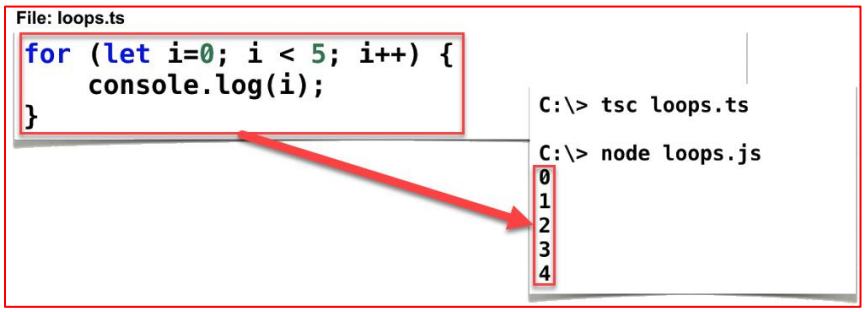
Creating Loops and Arrays – Overview

In TypeScript, we can make use of `for` loops, just like in other programming languages. For example, we have a file here named `loops.ts`, where we've set up a simple `for` loop:

```
for(let i = 0; i < 5; i++)
```

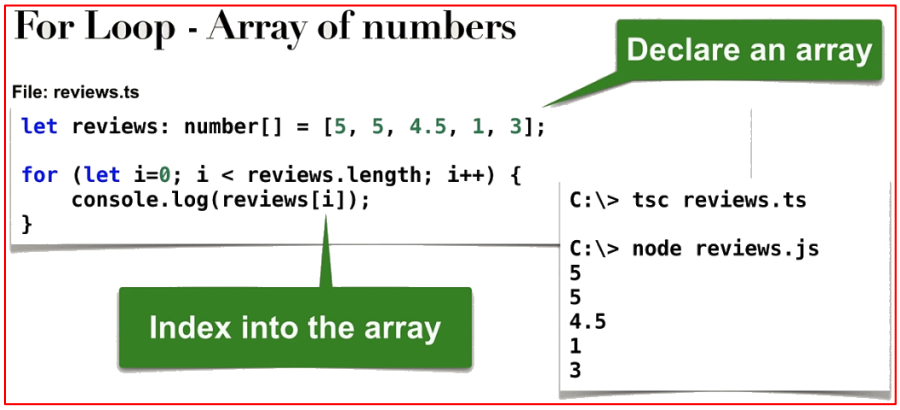


This loop will print out the value of `i` using `console.log(i)` each time it runs. To execute this code, we compile it using the `tsc` command, and then run the resulting JavaScript file using the `node` command. As expected, this will print out the numbers 0 through 4.

Now, let's see how we can use a `for` loop to iterate over an array of numbers. we'll declare an array named `reviews` with some initial values. Then, we'll set up a `for` loop to iterate over the array and print out each review. The loop looks like this:

```
for(let i = 0; i < reviews.length; i++)
```

, and to print each review, we'll use `console.log(reviews[i])`.



Next, let's use a `for` loop to perform some math operations. In this example, we'll compute the average of all the reviews. First, we'll initialize a variable `total` to 0. Then, inside the loop, we'll add each review to `total`. After the loop, I'll compute the average by dividing `total` by the number of reviews and then display the result. The output will be the average of all the reviews.

File: reviews.ts

```
let reviews: number[] = [5, 5, 4.5, 1, 3];

let total: number = 0;

for (let i=0; i < reviews.length; i++) {
    console.log(reviews[i]);
    total += reviews[i];
}

let average: number = total / reviews.length;

console.log("Review average = " + average);
```

```
C:\> tsc reviews.ts
C:\> node reviews.js
5
5
4.5
1
3
Review average = 3.7
```

Now, let's set up an array of strings. I'll create an array named `sports` and initialize it with some sports names. Then, I'll set up a `for` loop to print out each sport. Here, I'll also introduce a simplified `for` loop syntax that allows us to directly iterate over the elements of an array. Instead of using an index to access each element, we can directly access each element like this:

```
for(let tempSport of sports)
```

This loop will print out each sport in the `sports` array.

File: sports.ts

```
let sportsOne: string[] = ["Golf", "Cricket", "Tennis", "Swimming"];

for (let tempSport of sportsOne) {

    console.log(tempSport);
}
```

Current array element

```
C:\> tsc sports.ts

C:\> node sports.js
Golf
Cricket
Tennis
Swimming
```

We can also use conditionals inside our `for` loops. For example, if I want to highlight my favorite sport, I can add an `if` statement inside the loop to check if the current sport is my favorite, and if so, print a special message.

File: sports.ts

```
let sportsOne: string[] = ["Golf", "Cricket", "Tennis", "Swimming"];

for (let tempSport of sportsOne) {

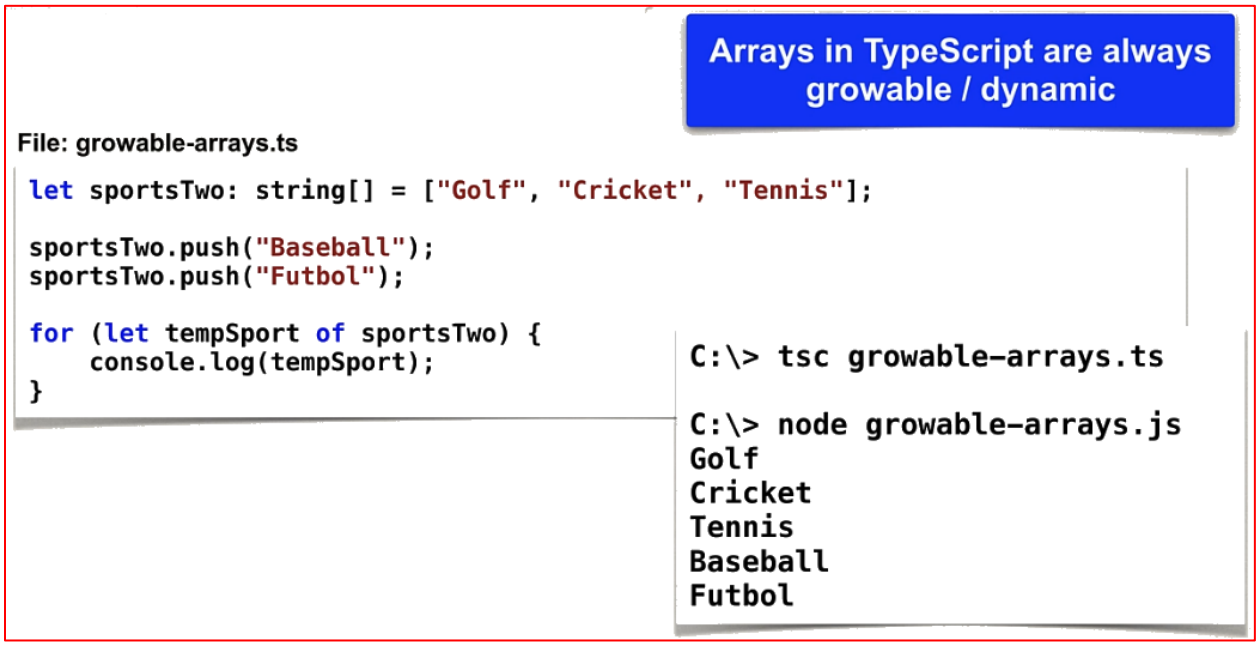
    if (tempSport == "Cricket") {
        console.log(tempSport + " << My Favorite!");
    }
    else {
        console.log(tempSport);
    }

}
```

```
C:\> tsc sports.ts

C:\> node sports.js
Golf
Cricket << My Favorite!
Tennis
Swimming
```

Next, let's discuss the nature of arrays in TypeScript. Arrays in TypeScript are always global and dynamic. This means that you can add elements to an array at any time, and the array will grow accordingly. For example, we can create an array `sports` with three elements, and then use the `push` method to add more elements to the array. Then, we can use a `for` loop to print out all the elements of the array.



In summary, we have learned how to use `for` loops in TypeScript to iterate over arrays, perform mathematical operations, and use conditionals inside loops. We have also learned about the dynamic nature of arrays in TypeScript.

Now, let's move into our IDE and write some code!

Creating Loops and Arrays – Write Some Code

In this section, we will discuss TypeScript loops and arrays in action. TypeScript allows us to use for loops, a fundamental concept familiar to most programmers. For example, consider a file named 'loops' in which we set up a basic for loop like this:

```
for (let i = 0; i < 5; i++) {
    console.log(i);
}
```

```
console.log('----- F O R  L O O P 1-------');
for (let i = 0; i < 5; i++) {
    console.log(i);
}
```

We can compile this TypeScript code using the `tsc` command and then execute the resulting JavaScript code with the `node` command. This will produce the expected output, just like in any other programming language.

In addition to basic loops, we can also use for loops to iterate over an array of numbers. Let's create an array called 'reviews' with some initial values:

```
let reviews: number[] = [5, 5, 4, 3, 2];
```

Now, we can set up a for loop to iterate over the 'reviews' array and print each review:

```
for (let i = 0; i < reviews.length; i++) {
    console.log(reviews[i]);
}
```

```
console.log('----- F O R  L O O P 2-------');
let reviews: number[] = [5, 5, 4, 3, 2];
for (let i = 0; i < reviews.length; i++) {
    console.log(reviews[i]);
}
```

This loop will print each review in the 'reviews' array. It is important to note that we use `reviews.length` to determine the number of iterations, and `reviews[i]` to access the individual elements of the array.

We can also perform mathematical operations inside a for loop. For example, we can compute the average of the 'reviews' array. First, we initialize a variable 'total' to zero. Then, inside the for loop, we add each review to the 'total':

```
let total = 0;
for (let i = 0; i < reviews.length; i++) {
    total += reviews[i];
}
let average = total / reviews.length;
console.log("Review average = " + average);
```

```
console.log('----- A V E R A G E -------');
let total = 0;
for (let i = 0; i < reviews.length; i++) {
    total += reviews[i];
}
let average = total / reviews.length;
console.log("Review average = " + average);
```

This will output the average of the 'reviews' array, which in this case is 3.7.

Next, let's set up an array of strings called 'sports':

```
let sports: string[] = ["Golf", "Cricket", "Tennis", "Swimming"];
```

We can use a traditional for loop with an index to print each sport:

```
for (let i = 0; i < sports.length; i++) {
    console.log(sports[i]);
}
```

```
let sports: string[] = ["Golf", "Cricket", "Tennis", "Swimming"];
for (let i = 0; i < sports.length; i++) {
    console.log(sports[i]);
}
```

However, TypeScript also offers a simplified for loop syntax that can be more readable:

```
for (let tempSport of sports) {
    console.log(tempSport);
}
```

```
for (let tempSport of sports) {
    console.log(tempSport);
}
```

This loop will print each sport in the 'sports' array without explicitly using an index.

We can also use conditionals inside our for loops. For example, we can modify the previous loop to indicate our favorite sport:

```
for (let tempSport of sports) {
    if (tempSport === "Cricket") {
        console.log(tempSport + " is my favorite!");
    } else {
        console.log(tempSport);
    }
}
```



This will print each sport, but when it reaches "Cricket," it will indicate that it is our favorite.

Arrays in TypeScript are always global and dynamic, meaning we can add elements to an array after its initial definition. For example, let's define an array 'sportsTwo' with three initial sports:

```
let sportsTwo: string[] = ["Golf", "Cricket", "Tennis"];
```

Now, we can add more sports to the 'sportsTwo' array using the `push` method:

```
sportsTwo.push("Baseball");
sportsTwo.push("Football");
```

Then, we can print each sport in the 'sportsTwo' array using a for loop:

```
for (let tempSport of sportsTwo) {
    console.log(tempSport);
}
```



This will print all five sports in the 'sportsTwo' array, including the ones we added later.